Light

GCSE Computer Science – Cyber Security Threats (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the general term for malicious software designed to damage, disrupt, or gain unauthorised access to computer systems?

- ○ **A.** Malware
- ○ **B.** Hardware
- ○ **C.** Firmware
- ○ **D.** Freeware

Q2 (1 mark)

What is a key characteristic of a computer virus?

- ○ **A.** It spreads automatically between computers without any user action
- ○ **B.** It requires a host file and user action to spread
- ○ **C.** It displays unwanted advertisements
- ○ **D.** It encrypts files and demands payment

Q3 (1 mark)

How does a computer worm differ from a virus?

- ○ **A.** A worm can only infect one computer
- ○ **B.** A worm requires a host file to operate
- ○ **C.** A worm can self-replicate and spread across networks without user action
- ○ **D.** A worm only affects mobile devices

Q4 (1 mark)

What is a Trojan horse in computer security?

- ○ **A.** Software that encrypts all your files
- ○ **B.** Software that automatically displays advertisements
- ○ **C.** Malware disguised as legitimate software to trick users into installing it

- ○ **D.** A physical device used to hack computers

Q5 (1 mark)

What is the primary purpose of ransomware?

- ○ **A.** To encrypt user files and demand payment for decryption
- ○ **B.** To display unwanted advertisements
- ○ **C.** To record user keystrokes
- ○ **D.** To slow down the computer processor

Q6 (1 mark)

What does spyware typically do?

- ○ **A.** Secretly monitors and collects user information without consent
- ○ **B.** Deletes all files on the computer
- ○ **C.** Prevents the computer from starting up
- ○ **D.** Encrypts files and demands money

Q7 (1 mark)

What is phishing?

- ○ **A.** A type of encryption used to protect data

- ○ **B.** A technique to speed up internet connections
- ○ **C.** A fraudulent attempt to obtain sensitive information by pretending to be a trustworthy source
- ○ **D.** A method to remove viruses from a computer

Q8 (1 mark)

What is social engineering in the context of cyber security?

- ○ **A.** Designing social media websites
- ○ **B.** Manipulating people into revealing confidential information or performing actions
- ○ **C.** Programming robots to interact with humans
- ○ **D.** Building secure network infrastructure

Q9 (1 mark)

What is a brute-force attack?

- ○ **A.** Systematically trying every possible password combination until the correct one is found
- ○ **B.** Physically breaking into a building to access computers
- ○ **C.** Sending millions of emails to crash a server
- ○ **D.** Tricking users into clicking malicious links

Q10 (1 mark)

What is the purpose of a Denial of Service (DoS) attack?

- ○ **A.** To steal user passwords
- ○ **B.** To encrypt files for ransom
- ○ **C.** To overwhelm a system with traffic so legitimate users cannot access it
- ○ **D.** To install spyware on computers

Q11 (1 mark)

What does SQL injection target?

- ○ **A.** The computer's operating system
- ○ **B.** Databases accessed through web applications
- ○ **C.** Email servers
- ○ **D.** Wireless network connections

Q12 (1 mark)

What is data interception?

- ○ **A.** Backing up data to a secure location
- ○ **B.** Encrypting data before transmission

- ○ **C.** Capturing data as it travels across a network without authorisation
- ○ **D.** Compressing data to save storage space

Q13 (2 marks)

A user receives an email that appears to be from their bank, asking them to click a link and verify their account details urgently.
Which features suggest this might be a phishing attempt?

- ○ **A.** The email creates urgency, requests sensitive information, and the link goes to an unfamiliar website address
- ○ **B.** The email contains the bank's official logo and uses formal language
- ○ **C.** The email was received during normal business hours
- ○ **D.** The email includes a phone number to call

Q14 (2 marks)

A company's network becomes infected with malware that spreads rapidly to hundreds of computers within hours, without any employees opening email attachments.
What type of malware is most likely responsible?

- ○ **A.** A virus, because it infected many computers
- ○ **B.** A Trojan, because it spread without users noticing
- ○ **C.** A worm, because it self-replicated across the network without user action
- ○ **D.** Ransomware, because it affected the entire company

Q15 (2 marks)

A user has a password that is 4 characters long using only lowercase letters (26 possibilities per character).
Why is this password vulnerable to brute-force attacks?

- ○ **A.** Because lowercase letters are easier to type than numbers
- ○ **B.** Because brute-force attacks only work on letters, not numbers
- ○ **C.** Because 4 characters is too long for attackers to guess
- ○ **D.** Because the short length and limited character set mean there are only 456,976 possible combinations, which can be tried very quickly

Q16 (2 marks)

What is the difference between a DoS attack and a DDoS attack?

- ○ **A.** DoS attacks come from a single source while DDoS attacks come from multiple distributed sources, making them harder to stop
- ○ **B.** DoS attacks steal data while DDoS attacks only disrupt services
- ○ **C.** DDoS attacks are legal while DoS attacks are illegal
- ○ **D.** DoS attacks target websites while DDoS attacks target email servers

Q17 (2 marks)

A website login form does not properly validate user input. An attacker enters malicious SQL code in the username field.
What could this SQL injection attack potentially achieve?

- ○ **A.** Speed up the website loading time
- ○ **B.** Encrypt the website's files
- ○ **C.** Slow down the user's internet connection
- ○ **D.** Bypass authentication, access or delete database records, or reveal sensitive user data

Q18 (2 marks)

Why are humans often described as the 'weak point' in computer security?

- ○ **A.** Because humans can be manipulated through social engineering to bypass even the strongest technical security measures
- ○ **B.** Because humans are physically weaker than computers
- ○ **C.** Because humans cannot remember complex passwords
- ○ **D.** Because humans work more slowly than computers

Q19 (2 marks)

A user downloads what appears to be a free game from an unofficial website. After installation, the attacker gains remote access to the computer.
What type of malware did the game contain?

- ○ **A.** A virus that spread from the game files
- ○ **B.** A worm that self-replicated across the network
- ○ **C.** Ransomware that encrypted the user's files

- ○ **D.** A Trojan that disguised malicious software as a legitimate game

Q20 (2 marks)

A business employee uses public Wi-Fi at a coffee shop to send sensitive company emails.
What security risk does this create?

- ○ **A.** The laptop battery will drain faster on public Wi-Fi
- ○ **B.** The coffee shop could charge for the data used
- ○ **C.** The emails will be permanently deleted from the server
- ○ **D.** Attackers on the same network could intercept unencrypted data as it travels through the network

Q21 (2 marks)

A hospital's computer systems are infected with ransomware. Critical patient records are now encrypted and inaccessible.
What are the main impacts of this attack?

- ○ **A.** Only the hospital's website will be affected
- ○ **B.** Patient care could be disrupted, lives potentially endangered, and the hospital faces pressure to pay ransom to restore access
- ○ **C.** The hospital's phone lines will stop working
- ○ **D.** Staff members' personal bank accounts will be emptied

Q22 (2 marks)

A caller claims to be from the IT department and asks an employee for their password to 'fix a security issue'.
What type of attack is this?

- ○ **A.** A brute-force attack on the phone system
- ○ **B.** A denial of service attack on the employee
- ○ **C.** A SQL injection through the phone line
- ○ **D.** A social engineering attack using pretexting (creating a false scenario) to obtain credentials

Q23 (2 marks)

After visiting an untrusted website, a user notices their bank account has been accessed. Investigation reveals software was secretly recording everything typed on the keyboard.
What type of malware was installed?

- ○ **A.** A worm that spread to the bank's systems
- ○ **B.** A keylogger (type of spyware) that captured login credentials as they were typed
- ○ **C.** Ransomware that locked the bank account
- ○ **D.** A Trojan that modified bank transactions

Q24 (3 marks)

A company experiences the following events: employees receive urgent emails appearing to be from the CEO requesting gift card purchases, the company website crashes from overwhelming traffic, and customer data is found for sale online.
Analyse these incidents and identify the types of attacks that have occurred.

- ○ **A.** All three are examples of virus infections spreading through the network
- ○ **B.** All three are caused by a single piece of ransomware
- ○ **C.** The incidents are unrelated technical failures, not attacks
- ○ **D.** The urgent emails are phishing (social engineering), the website crash is a DoS/DDoS attack, and the data sale indicates data theft (possibly from earlier compromise)

Q25 (3 marks)

A security analyst needs to explain the differences between viruses, worms, and Trojans to train new staff.
Analyse the key distinguishing characteristics of each malware type.

- ○ **A.** They are all the same thing with different names
- ○ **B.** Viruses require host files and user action to spread; worms self-replicate across networks without user action; Trojans disguise themselves as legitimate software to trick users into installation
- ○ **C.** Viruses affect only Windows, worms affect only Mac, and Trojans affect only Linux
- ○ **D.** All three require user permission before they can cause any damage

Q26 (3 marks)

A web developer's login form is vulnerable to SQL injection. When a user enters their username, it is directly inserted into the SQL query without validation.
Analyse how an attacker could exploit this and what damage they could cause.

- ○ **A.** The attacker could only view the login page design
- ○ **B.** The attacker could only slow down the website
- ○ **C.** SQL injection cannot cause any real damage to databases
- ○ **D.** The attacker could enter SQL commands like 'OR 1=1' to bypass login, then potentially view all user records, extract sensitive data, modify or delete database contents, or escalate privileges

Q27 (3 marks)

A major online retailer is hit by a DDoS attack during their biggest sale event. The attack uses 100,000 compromised computers worldwide. Analyse why this type of attack is particularly difficult to defend against.

- ○ **A.** DDoS attacks are easy to stop because all traffic comes from one location
- ○ **B.** The distributed nature means traffic comes from many legitimate-looking sources, blocking one source has minimal impact, and distinguishing attack traffic from genuine customer traffic during a sale is extremely challenging
- ○ **C.** DDoS attacks only affect email systems, not websites
- ○ **D.** Modern websites are completely immune to DDoS attacks

Q28 (3 marks)

An attacker researches a company's finance director on social media, noting their recent holiday photos and children's school events. The attacker then sends a targeted email mentioning these personal details. Analyse why this attack is more dangerous than generic phishing.

- ○ **A.** It is actually less dangerous because it only targets one person

- ○ **B.** Social media research does not help attackers in any way
- ○ **C.** Finance directors cannot be targets of phishing attacks
- ○ **D.** The personalised details make the email seem more credible and trustworthy; targeting a high-value individual with authority over finances increases potential damage; and the research demonstrates sophisticated, determined attackers

Q29 (3 marks)

A school's network is infected with ransomware. All student records, grades, and teaching materials are encrypted. The attackers demand payment in cryptocurrency.
Analyse the options available to the school and the implications of each.

- ○ **A.** The only option is to pay the ransom immediately
- ○ **B.** Options include: restore from backups (if available and uninfected), rebuild systems and accept data loss, pay ransom (risky with no guarantee), or engage law enforcement and cybersecurity specialists to explore recovery methods
- ○ **C.** The school should ignore the problem and continue using paper records
- ○ **D.** Ransomware always provides working decryption keys after payment

Q30 (3 marks)

An attacker sends phishing emails to company employees. One employee clicks a link and unknowingly installs a keylogger. The keylogger captures the IT administrator's credentials when they log in on that computer. The attacker then uses these credentials to access the company network and install ransomware on all servers.
Analyse how multiple attack types were chained together in this incident.

- ○ **A.** Only one type of attack was used throughout
- ○ **B.** The attacks were completely separate and unrelated incidents
- ○ **C.** Keyloggers cannot capture administrator passwords
- ○ **D.** The chain used: phishing (initial access via deceptive email), spyware/keylogger (credential theft), and ransomware (final payload) - each attack enabled the next, escalating from one compromised user to organisation-wide impact

Total: **55** marks • Time: **30 min**

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